Stochastic Dynamics of Multi-Agent RL

From population dynamics to populations of learners

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Iliad Intensive · Multi-Agent AI day · October 2026

Plan for today

AI systems increasingly learn alongside each other.
What happens to the population?

Aim: model a population of learners as a dynamical system, and see where such models could matter for AI safety.

1. Why model multi-agent systems?
2. Background: population dynamics, MDPs, RL
3. Case study: Chu et al. (AAAI 2023)
4. Extension: Geng, Barfuss & Chen (AAAI 2026)
5. Where next?

Part 1

Why model multi-agent systems?

AI is already a multi-agent world

In deployment

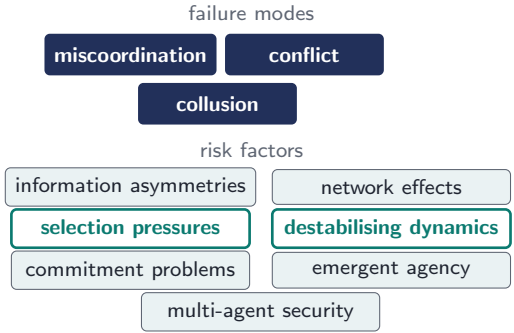
- Economies of autonomous AI agents
- Moltbook: a social network used only by AI agents

In training

- GPT-Red: an attacker trained against a *population* of defenders

Safety doesn't add up

- "A collection of safe agents does not guarantee a safe collection of agents"

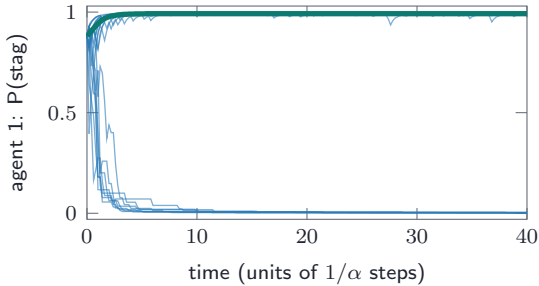


Taxonomy of Hammond et al. (2025)

Tomašev et al. 2025 · De Marzo et al. 2026 · Wallace et al. 2026 · Gradient Institute 2025 · Hammond et al. 2025

Why build models?

- Learning need not settle at an equilibrium: it can be **chaotic**
- Small details of the update rule change the outcome
- Simulations are expensive and hard to interpret
- Noise can change *which* outcome you reach



Two Q-learners, stag hunt, same start (our simulation). Mean equation (thick): stag. 8/20 runs: hare.

Galla & Farmer 2013 · Sanders et al. 2018 · Asker et al. 2022 · Barfuss et al. 2025 · Barfuss & Meylahn 2023

Part 2

Background

Population dynamics

Track the **frequencies** of types, not the individuals.

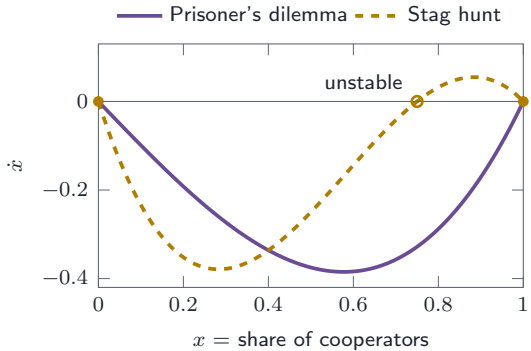
Replicator equation

$$\dot{x}_i = x_i\big[f_i(x) - \bar{f}(x)\big]$$

Types doing better than average grow.
Infinite, well-mixed population.

Finite populations

Moran and Wright–Fisher processes: drift, fixation, diffusion limits.

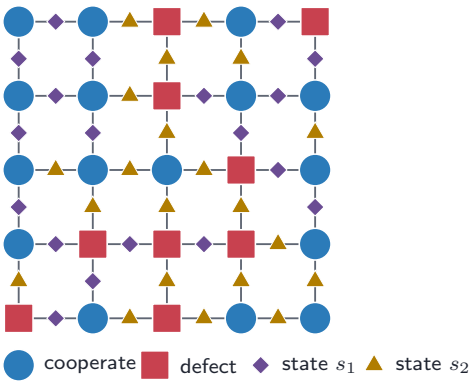


PD: everything flows to defection. Stag hunt: two basins.

Taylor & Jonker 1978 · Hofbauer & Sigmund 1998

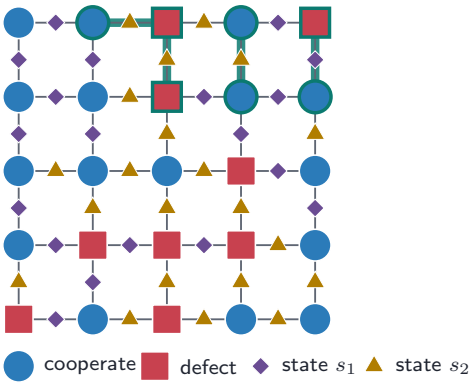
Mean field on a network

Agents on a network. Each link is a game, played in state s_1 or s_2.



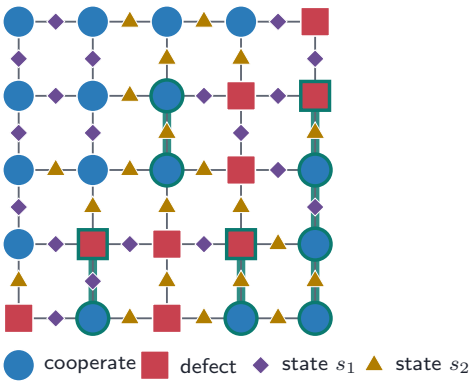
Mean field on a network

Linked pairs play their games.



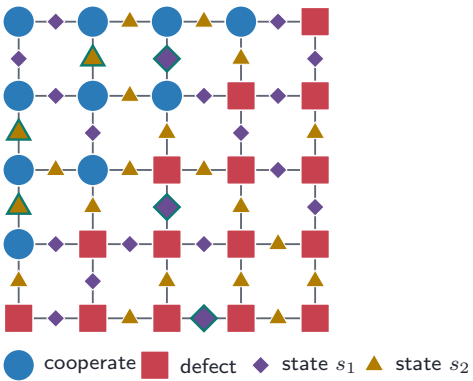
Mean field on a network

Those agents update their strategies; the next pairs play.



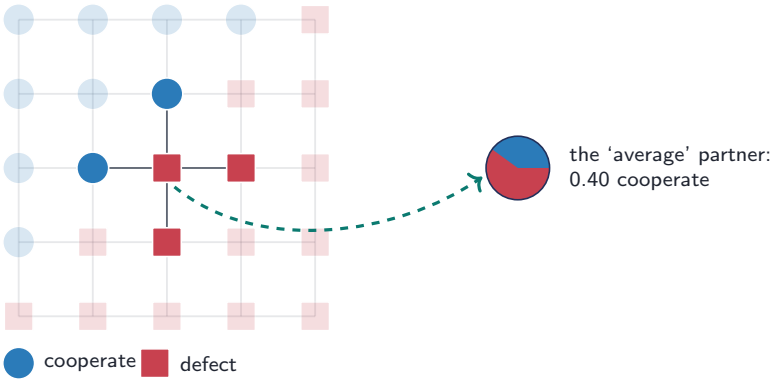
Mean field on a network

More updates, and some game states change: cooperators cluster.



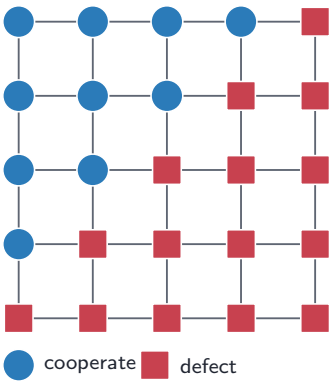
Mean field on a network

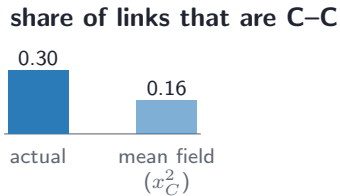
Mean field: each agent plays against the population *average*.



Mean field on a network

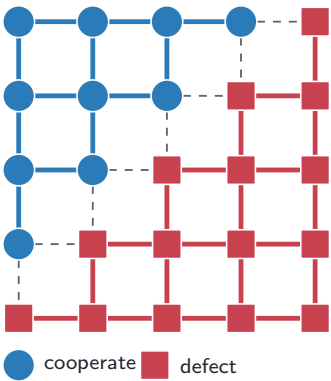
But neighbours are correlated: mean field gets the C–C links wrong.

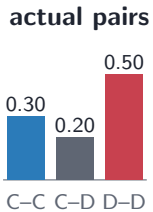


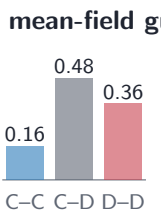


Pair approximation

Pair approximation: track the frequencies of neighbouring *pairs*.



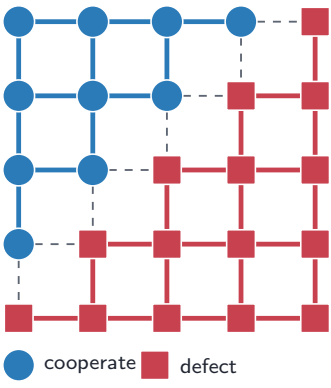


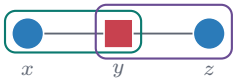


Hauert & Szabó, Am. J. Phys. 2005 · Szabó & Fáth, Phys. Rep. 2007

Pair approximation

Close the hierarchy: rebuild triples from pairs.



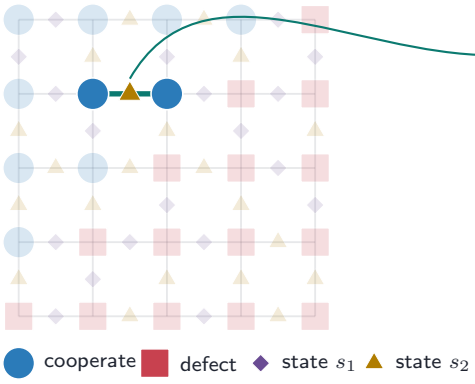


$$p(x,y,z) \approx \frac{p(x,y)\,p(y,z)}{p(y)}$$

Hauert & Szabó, Am. J. Phys. 2005 · Szabó & Fáth, Phys. Rep. 2007

Pair approximation

Next: a node's type becomes its Q-values, and a link carries a game state.



$\langle Q^1,\ s,\ Q^2 \rangle$

node: its Q-values
link: the state of the pair's game

Hauert & Szabó, Am. J. Phys. 2005 · Szabó & Fáth, Phys. Rep. 2007

A classic result from pair approximation

On a graph where everyone has k neighbours,

cooperation is favoured when

$$\frac{b}{c} > k$$

b = benefit given, c = cost paid

Conditions: weak selection, death–birth updating, $k \ll N$.

Ohtsuki, Hauert, Lieberman & Nowak, Nature 2006

MDP and RL refresher

- MDP: $(S, A, T(s' \mid s, a), R(s, a, s'), \gamma)$, policy $\pi(a \mid s)$
- Q-learning:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\big[r + \gamma \max_{a'} Q(s',a') - Q(s,a)\big]$$

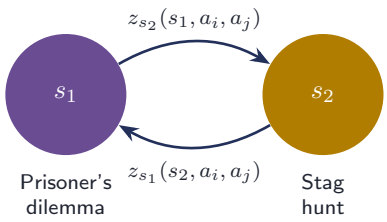
- **New today:** Boltzmann exploration

$$x(a) = \frac{e^{\tau Q(a)}}{\sum_{a'} e^{\tau Q(a')}}$$

$\tau = 0$: uniformly random. $\tau \to \infty$: greedy.

Stochastic games

- State transitions and rewards depend on the *joint* action
- Other learners are part of my environment, so it keeps changing
- Single-agent convergence guarantees no longer apply
- **Myopic, stateless learner:** one Q-value per action, $\gamma = 0$



The pair's joint action sets the chance of each move.

Shapley, PNAS 1953 · Zhang, Yang & Başar, "Multi-agent RL: a selective overview"

Why learning and population dynamics are linked

Cross learning $\rightarrow$ replicator dynamics

Boltzmann Q-learning $\rightarrow$ selection + mutation

$$\dot{x}_i = \underbrace{\alpha\tau\, x_i\big[(Ay)_i - x^\top Ay\big]}_{\text{selection (replicator)}} + \underbrace{\alpha\, x_i \sum_j x_j \ln \frac{x_j}{x_i}}_{\text{mutation (exploration)}}$$

evolution	learning
type	action
fitness	reward
mutation	exploration

Rest points: $x_i \propto e^{\tau(Ay)_i}$

a logit QRE, **not Nash**

Börgers & Sarin, JET 1997 · Tuyls, Verbeeck & Lenaerts, AAMAS 2003

Where the selection–mutation equation comes from

1. Expected update for *every* action

$$\dot{Q}_i = \alpha \left(r_i + \gamma \max_k Q_k - Q_i \right)$$

Tuyls, Verbeeck & Lenaerts, AAMAS 2003 · Kaisers & Tuyls, AAMAS 2010 · Bloembergen et al., JAIR 2015

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2. Chain rule through the softmax $x_i = e^{\tau Q_i} / \sum_k e^{\tau Q_k}$

$$\frac{\dot{x}_i}{x_i} = \tau \Big(\dot{Q}_i - \sum_j x_j \dot{Q}_j \Big) \qquad (\gamma \max Q \text{ cancels})$$

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Step 1 is an idealisation: standard Q-learning only updates the action it took.
Frequency-adjusted Q-learning corrects for this.

Tuyls, Verbeeck & Lenaerts, AAMAS 2003 · Kaisers & Tuyls, AAMAS 2010 · Bloembergen et al., JAIR 2015

From two learners to infinitely many

- Infinite, well-mixed population of Q-learners (mean field)
- The object becomes a **distribution over Q-values**, $p(Q, t)$
- Extensions: a continuity-equation model; regular graphs

Open question

What if each interaction has a *state* that depends on who you are paired with and what you both did?

Hu, Leung & Leung, NeurIPS 2019 · Hu, Leung, Leung & Soh, AAMAS 2022 · Chu et al., IJCAI 2022

Part 3

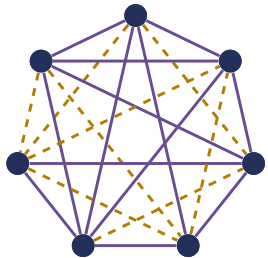
Case study: Chu et al. 2023

The setting

- n Q-learners, complete graph; theory takes $n \to \infty$
- Myopic ($\gamma = 0$) and blind to the state
- Each link: a two-player stochastic game, symmetric transitions

Each step

1. Sample *one* action, play it against everyone
2. Reward = average over $n - 1$ games
3. Update only that action's Q-value
4. Each link's state moves via $z_s(s', a_i, a_j)$



Solid: state s_1. Dashed: state s_2.

One agent's velocity

$$v_j(Q, a_j) = \alpha \left(\mathbb{E}[r(a_j)] - Q(a_j) \right) \qquad \text{(eq. 5)}$$

- $\mathbb{E}[r(a_j)]$ depends on my **state distribution** and on whom I meet
- Infinitely many opponents: reward noise averages out
- Still random: *which action* is sampled

Why mean field fails here

Mean field: play against the population's average strategy $\bar{x}$

$$\mathbb{E}[r(a_j)] = e_j^\top M\bar{x} \qquad \text{(eq. 6)}$$

With states:

- My actions shape the states of *my own* games
- So my state distribution, and the strategies I meet, depend on *my* Q-values
- Agents feel different "fields": mean field assumes they all feel the same one

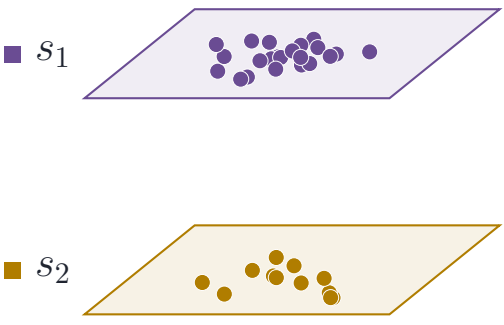
Pairs of Q-learners

A **pair** is $\langle Q^1, s, Q^2 \rangle$:
my Q-values, our game's state, your Q-values

Track its density and split it (eq. 7):

$$p(Q^1, s, Q^2, t) = p(s, t)\; p(Q^1, Q^2, t \mid s)$$

This gives each agent's state distribution and the strategies it meets.



Derivation, step 1: the states

Share of pairs in state s

$$p(s,t) = \iint p(Q^1, s, Q^2, t)\, \mathrm{d}Q^1 \mathrm{d}Q^2 \qquad \text{(eq. 8)}$$

How it changes

$$\frac{\mathrm{d}p(s,t)}{\mathrm{d}t} = \sum_{s'} \iint p(Q^1, s', Q^2, t) \sum_{i,j} x_i(Q^1)\, x_j(Q^2)\, z_s(s', a_i, a_j)\, \mathrm{d}Q^1 \mathrm{d}Q^2 \; - \; p(s,t)$$

- Inflow: pairs in any state s' that play (a_i, a_j) and move to s
- Outflow: everything currently in s (some of it returns via $s' = s$)

Chu et al., AAAI 2023, eq. (10), following Tuyls, Verbeeck & Lenaerts 2003

Derivation, step 2: freeze the states

Within a step, learning and state transitions are independent, so first move the Q-values with the states held fixed.

Master equation in Q-space $P = (Q^1, Q^2)$

$$\frac{\partial p^s(P,t)}{\partial t} = \int \left[T(P,P')\, p^s(P',t) - T(P',P)\, p^s(P,t) \right] \mathrm{d}P' \qquad \text{(eq. 11)}$$

Chu et al., AAAI 2023, eqs. (11)–(12), supplement S.1–S.4. Written here in conservative form; the paper prints eq. (12) differently.

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A pair reaches P from $P' = P - v\,\Delta t$ only via joint action (a_i, a_j), with probability $x_i\, x_j$. Taylor-expand around P:

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A pair reaches P from $P' = P - v\,\Delta t$ only via joint action (a_i, a_j), with probability $x_i\, x_j$. Taylor-expand around P:

$$\frac{\partial p^s}{\partial t} = -\sum_i \frac{\partial}{\partial Q^1(a_i)} \Big[v_i\, x_i(Q^1)\, p^s \Big] \;-\; \sum_j \frac{\partial}{\partial Q^2(a_j)} \Big[v_j\, x_j(Q^2)\, p^s \Big]$$

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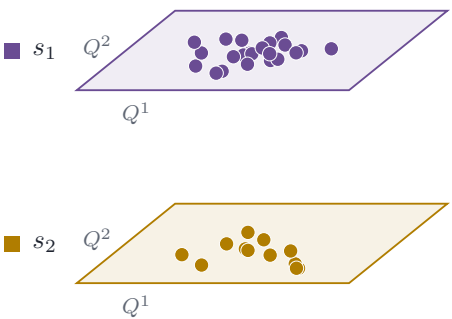
Derivation, step 3: put the transitions back

After the flow in Q-space, **move mass between the state planes**:

$$\frac{\partial p(P,t\,|\,s)}{\partial t} \;=\; \frac{1}{p(s,t)}\Big[\; \underbrace{\textstyle\sum_{s'}(\ldots)\,x_i\,x_j\,z_s(s',a_i,a_j)}_{\text{mass arriving in } s} \;-\; p(P,t\,|\,s)\,\frac{\mathrm{d}p(s,t)}{\mathrm{d}t}\Big] \qquad \text{(eq. 13)}$$

Read-out: integrate the pair density (eq. 14)

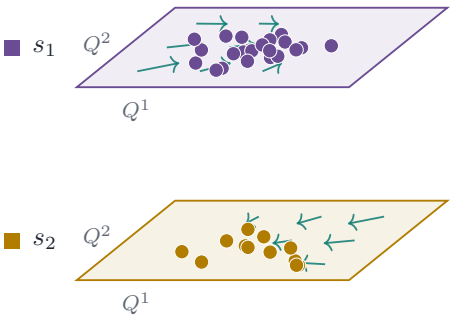
Putting the pieces together



Each dot is one pair (Q^1, Q^2), drawn on the plane of its game state.

Schematic: toy dynamics showing the structure of the model, not results from the paper.

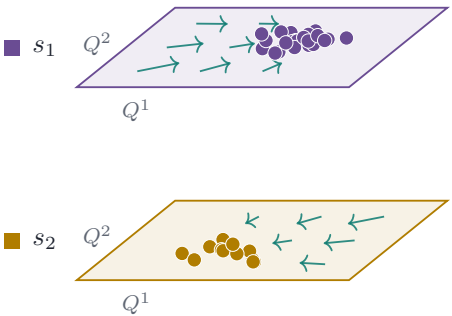
Putting the pieces together



Learning moves mass *within* a plane, along the velocity field.

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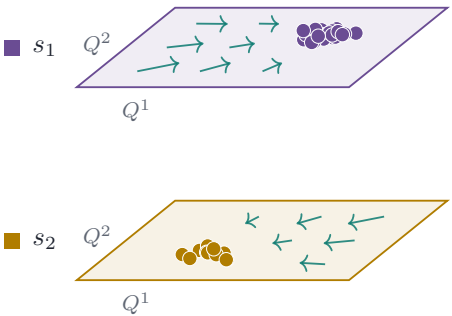
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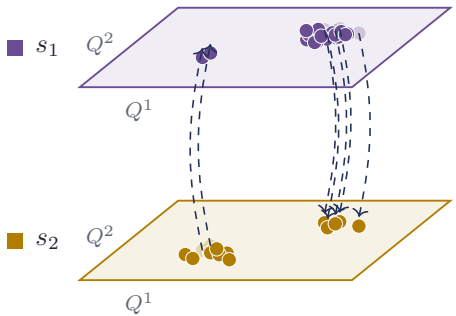
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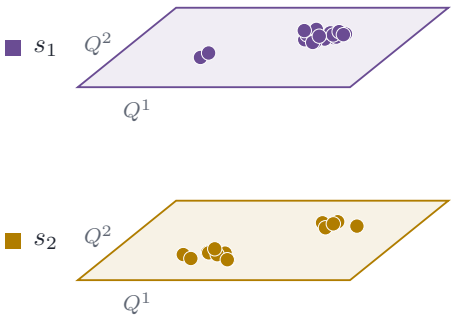
Putting the pieces together



State transitions move mass *between* planes, at rates set by the joint action.

Schematic: toy dynamics showing the structure of the model, not results from the paper.

Putting the pieces together



Read-out: integrate the density over time

fraction of pairs in s_1



average Q^1 of pairs in s_1



Schematic: toy dynamics showing the structure of the model, not results from the paper.

Results

- Theory matches simulation across starting points, transition rules, α and τ
- Holds on average even at $n = 10$, though single runs vary a lot
- State transitions **promote cooperation**, even for myopic learners

Chu et al., AAAI 2023, and supplement

What's missing?

- Agents are myopic and ignore the state
- Everyone meets everyone (complete graph)
- No diffusion term: run-to-run noise is not modelled
- The equation lives in $2m$ dimensions, for each state
- Transitions must be symmetric

Part 4

Extension: Geng et al. 2026

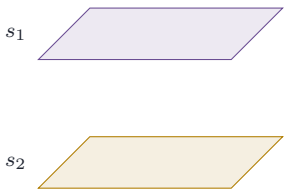
What changed, and why

	Chu et al. 2023	**Geng et al. 2026**
Learner	myopic, one Q per action	Q per state and action, $\gamma > 0$
Structure	complete graph	lattice; random regular graphs
Closure	pairs $\langle Q^1, s, Q^2 \rangle$	2nd neighbour $\approx$ 1st neighbour
Fluctuations	not represented	SDE per agent $\to$ Fokker–Planck
Reduction	none	separation of timescales

Why: richer learners, local structure, a test for when noise matters, a smaller model.

Fast states, slow learning

$$\partial_t p(Q, s, \tilde{Q}) \;=\; \underbrace{\text{state transitions}}_{O(1)} \;-\; \underbrace{\nabla\cdot(\mu\, p)}_{\text{drift},\, O(\alpha)} \;+\; \underbrace{\tfrac{1}{2}\nabla\nabla:(\Sigma\, p)}_{\text{diffusion},\, O(\alpha^2)}$$



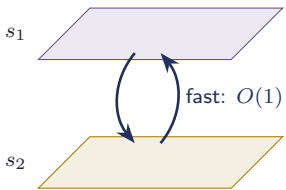
Geng, Barfuss & Chen, AAAI 2026, eqs. (15)–(19) · Kerzreho, arXiv:2506.21079

Extension: Geng et al. 2026

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- For $\alpha \ll 1$, states relax fast to $p^*(s \,|\, Q, \tilde{Q})$



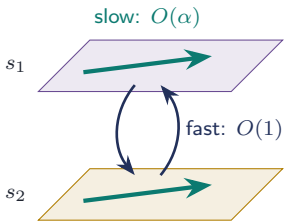
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- Q-values move slowly, as if the environment were fixed



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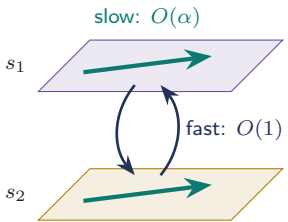
Extension: Geng et al. 2026

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- For $\alpha \ll 1$, states relax fast to $p^*(s \,|\, Q, \tilde{Q})$
- Q-values move slowly, as if the environment were fixed
- Drop $O(\alpha^2)$: each agent follows a **deterministic** path

$$\partial_t p(Q, \tilde{Q}) = -\sum_{F \in \{Q, \tilde{Q}\}} \nabla_F \cdot \left[\mu^*(F)\, p\right]$$



Geng, Barfuss & Chen, AAAI 2026, eqs. (15)–(19) · Kerzreho, arXiv:2506.21079

Extension: Geng et al. 2026

Results and follow-ups

- Two-state PD on a lattice: full cooperation in *both* states
- Theory fits simulation when $\alpha < 10^{-3}$
- More neighbours act like a **bigger batch**: faster learning

Follow-ups

- Actor-critic: chaos; several stable outcomes
- Networks: degree helps or hurts, by transition rule
- Partner selection: **population variance is necessary**

Geng et al., AAAI 2026 · Geng et al., arXiv:2601.07142 · Yuan et al., IEEE TNNLS 2026 · Russell, Leung & Turrini, arXiv:2605.18185

Extension: Geng et al. 2026

Part 5

Where next?

Cooperative AI: from MARL social dilemmas...

MARL roots

- Sequential social dilemmas · inequity aversion · LOLA · Melting Pot

Researchers who started with MARL

- **Elizaveta Tennant**: MARL agents with moral rewards
- **Lewis Hammond**: MARL with temporal logic; now at the Cooperative AI Foundation
- **Warwick**: Q-learners learn to leave defectors (Out-for-Tat)

Leibo et al. 2017, 2021 · Hughes et al. 2018 · Foerster et al. 2018 · Tennant et al., IJCAI 2023; AIES 2024 · Hammond et al., AAMAS 2021 · Leung et al., IJCAI 2024 · Russell et al., AAMAS 2026

. . . to LLM agents in the same games

- **Tennant**: fine-tune an LLM on the iterated PD with moral rewards
- **Hammond**: multi-agent risks; secret collusion between LLMs
- LLMs do well in the iterated PD, poorly in coordination games
- Modern LLMs **default to defection**; contracts and mediation help
- o3 cooperates less than o3-mini, even at zero cost

Same games, new agents, mostly studied by experiment.

Tennant et al., ICLR 2025 · Motwani et al. 2024 · Akata et al. 2025 · Piatti et al. 2024 · Tewolde et al., arXiv:2604.15267 · Yadav et al., MATS 2026

Three directions

How can classical, analytical theory help with current AI safety problems?

1. Inside training

Learning dynamics in training and evaluation

2. Safety-relevant MARL

MARL theory where safety is at stake

3. Populations of LLMs

Theory for many interacting LLMs: very early

1. The maths is already inside training

- RLHF optimum: $\pi^* \propto \pi_{\text{ref}}\, e^{r/\beta}$, a softmax response with a prior
- Nash learning from human feedback: $\pi_{t+1} \propto \pi_t^{\mu}\, e^{\eta P(y \succ \pi_t^{\mu})}$, $\pi_t^{\mu} \propto \pi_t^{1-\eta\tau} \mu^{\eta\tau}$

Korbak et al. 2022 · Munos et al., ICML 2024 · Liu et al., arXiv:2410.23223 · Perolat et al., Science 2022 · Irving et al. 2018

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Small-step limit (our derivation)

$$\frac{\dot{\pi}_y}{\pi_y} = \eta \Big[\underbrace{P(y \succ \pi) - P(\pi \succ \pi)}_{\text{selection}} \Big] - \eta\tau \Big[\underbrace{\ln \frac{\pi_y}{\mu_y} - \mathrm{KL}(\pi \,\|\, \mu)}_{\text{mutation towards } \mu} \Big]$$

Same structure as Boltzmann Q-learning (selection + mutation).

Korbak et al. 2022 · Munos et al., ICML 2024 · Liu et al., arXiv:2410.23223 · Perolat et al., Science 2022 · Irving et al. 2018

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Same structure as Boltzmann Q-learning (selection + mutation).

- Multiplicative-weights methods can diverge; regularised ones converge
- DeepNash: replicator dynamics on a regularised game
- Debate assumes optimal play: does training reach it?

Korbak et al. 2022 · Munos et al., ICML 2024 · Liu et al., arXiv:2410.23223 · Perolat et al., Science 2022 · Irving et al. 2018

2. Algorithmic collusion

1. Q-learning pricing agents learn above-competitive prices
2. A continuous-time limit explains it: "spontaneous coupling"
3. German petrol: margins +28% only if *both* stations use algorithms
4. LLM pricing agents do it too, and prompt wording matters
5. More sellers or mixed algorithms break it up; one shared model gives two stable price levels

What matters: population size, heterogeneity, timescales, finite-sample noise.

Calvano et al., AER 2020 · Banchio & Mantegazza, arXiv:2202.05946 · Assad et al., JPE 2024 · Fish et al., arXiv:2404.00806 · Keppo et al., arXiv:2603.20281 · Cao & Hu, arXiv:2601.01279

3. Populations of LLMs today

- Coordination game: runs turn deterministic above a critical size
- Conventions form; group bias without biased individuals
- Chu et al.'s group: reputation keeps LLM agents cooperating

How people get tractable theory today

1. Reduce agents to a few types, then use evolutionary game theory
2. Measure response probabilities, then treat agents as interacting particles
3. Treat repeated retraining as a Markov chain

Flint et al., arXiv:2510.22422 · Ashery et al., Sci. Adv. 2025 · El et al., arXiv:2608.16578 · Ren et al., arXiv:2505.05029

3. Open avenues (speculative)

- **Personas as types?** They seem context-dependent: like *state-dependent* policies
- **In-context learning** as the fast timescale, weight updates as the slow one
- **Partner selection** among LLM agents

classical	LLM version
types, Q-values	personas
state	context
fast state mixing	in-context adaptation
slow Q drift	weight updates
rewiring	partner selection

Analogies, not results

Anthropic, "The persona selection model", Feb 2026 · Marks, LessWrong, Sep 2026 · Nasser et al., arXiv:2602.16301 · Russell et al., arXiv:2605.18185

Summary

1. A population of learners is a **dynamical system**
2. Pair approximation and separation of timescales make stochastic games tractable
3. The same maths already sits inside frontier AI;
   populations of LLMs are largely unexplored

Thank you. Questions?

James Sykes · University of Warwick

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